

Letter of Recommendation for Xuan Thanh Trinh

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To: Ph.D. Selection Committee

Doctoral Program in Electrical, Electronics and Computer Engineering

College of Engineering and Computer Science (CECS)

University of Michigan–Dearborn, MI 48128, USA

Dear Admissions Committee,

I am Dr. Tran Thanh Son, a Lecturer in the School of Electrical and Electronic Engineering at Hanoi University of Science and Technology (HUST). As an expert in power system optimization and microgrid architectures (Ph.D., Shibaura Institute of Technology), I served as Xuan Thanh Trinh's direct research mentor on a complex project regarding peer-to-peer energy trading frameworks. Based on his exceptional performance, I strongly recommend him for the Ph.D. program at the University of Michigan–Dearborn.

Thanh demonstrates a rare capacity for mathematical abstraction and independent research execution. Rather than passively applying formulas, he rigorously evaluates their fundamental analytical properties before deployment. Notably, he independently established the global convergence proof for an ADMM framework combined with Second-Order Cone Programming (SOCP) to accommodate complex, non-linear grid constraints. As an independent researcher, he consistently navigates dense academic literature to master theoretical frameworks outside the undergraduate curriculum, requiring minimal direct supervision. This dual capability, profound mathematical rigor coupled with exceptional self-direction, proving his readiness for independent Ph.D. research.

Furthermore, Thanh exhibits exceptional system-level thinking and validation methodology. He independently designs comprehensive scenarios analyzing the highly non-linear physical coupling between autonomous microgrids and the utility network, particularly under high penetrations of distributed renewables and energy storage. To validate these complex system formulations, he subjects his algorithms to severe stress testing across both radial and meshed topologies such as the IEEE 123-bus system to successfully resolving critical ADMM nonconvergence via robust dynamic penalty mechanisms. His ability to anchor abstract system-level designs with realistic grid constraints provided the critical empirical foundation for our manuscript, currently under review at the journal Sustainable Energy, Grids and Networks (SEGAN).

Ultimately, Thanh's methodological resilience sets him apart during complex simulations. When debugging large-scale grid networks, he constructively treats algorithmic non-convergence as diagnostic data rather than failures. Even under strict deadlines, he systematically traces numerical instabilities to their theoretical root causes rather than relying on unverified heuristics. This analytical endurance proves he is exceptionally well-equipped for the rigor of a Ph.D. program.

In summary, Thanh uniquely combines mathematical rigor, system-level thinking, and methodological resilience. As his mentor and co-author, I give him my highest recommendation for admission to your Ph.D. program without reservation. I am confident he will be an immediate and highly productive contributor to your research group. Please contact me if you require further information. Sincerely,

Dr. Tran Thanh Son

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